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A palaeomagnetic study of Jurassic rocks from the Antarctic Peninsula and its implications

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SUMMARY: Palaeomagnetic measurements have been made on rocks of Jurassic age from the Antarctic Peninsula to determine whether there has been relative movement between the region and East Antarctica. Samples were collected from the Oscar II Coast and the Cole Peninsula. Reliable results have been obtained from Jurassic dykes dated in one area at 174 ± 5 Ma, from an acid pluton of approximately the same age and from lavas of middle Jurassic age. Combining these results a position for the palaeopole was obtained at 48°S 238°E. If a reconstruction of Gondwana consistent with sea-floor spreading, palaeomagnetic and geological data is adopted, such as that of Norton & Sclater (1979), then a well-determined position for the Antarctic Peninsula within the reconstruction is obtained. Space considerations suggest that during the Jurassic the relative positions of the Antarctic Peninsula and the Ellsworth block were not very different from the present day, though palaeomagnetic measurements (Watts & Bramall 1981) indicates that the orientation was different.

The palaeomagnetic work described below was undertaken to provide information about the relative positions of East and West Antarctica prior to the break-up of Gondwana, thus helping towards an understanding of the evolution of the Antarctic Peninsula and placing some constraint on possible reconstructions of the southern continents, about which there is as yet no complete agreement. Although some marine magnetic information is available about the relative motion of East Antarctica and Africa during the early stages of separation (Bergh 1977; Segoufin 1978; Simpson *et al.* 1979), it is not yet sufficient to make possible an unambiguous Jurassic reconstruction. This is reflected in the considerably differing models proposed by Norton & Sclater (1979), Powell *et al.* (1980) and Barron *et al.* (1978). There is now a very considerable body of palaeomagnetic data for the southern continents against which any reconstruction can be tested, though some workers are dubious about its value for this purpose. We shall, however, show that the statistical evidence points to a preference for the Norton & Sclater model. However, unlike the models of Powell *et al.* and Barron *et al.*, the Norton & Sclater model and similar reconstructions such as that of Smith & Hallam (1970) imply a relative movement between East Antarctica and the Peninsula, since without this the reconstruction produces a considerable overlap of the Peninsula and the Falkland Plateau, the latter known to be a continental area (Barker, Dalziel *et al.* 1976).

The Peninsula (in part at least Palaeozoic in age) is therefore a critical area. Some early palaeomagnetic measurements (e.g. Blundell 1962) were made on undated collections, mainly on rocks of the Andean Intrusive Suite, now known to cover ages from <50 Ma to >200 Ma (Pankhurst 1982). Measurements on younger rocks include those on lavas from the South Shetland Is known to be of Cenozoic age (Scharon *et al.* 1970; Valencio & Fourcade 1969; Dalziel *et al.* 1973; Watts 1982). Other collections include measurements on rocks as old as the Cretaceous (Valencio *et al.* 1979; Scharnberger & Scharon 1982; Kellogg & Reynolds 1978; Kellogg 1980). Prior to the present study no reliable palaeomagnetic measurements on properly dated Jurassic material had been made. The available palaeomagnetic measurements on younger rocks did not provide any conclusive evidence for or against relative movement of the Peninsula since the early Cretaceous. This is perhaps not surprising since the period under consideration is one of general drift of the southern continents, in addition to which, given the proximity to the South Pole, the relative movement sought is near the limit of resolution of the method.

Despite the general paucity of palaeomagnetic measurements from East Antarctica, the middle Jurassic pole there is well determined, largely because the initial rifting was accompanied by widespread doleritic intrusion. The data from which the East Antarctic palaeopole has been calculated are given in Table 1. Measurements made by Beck (1972) on the Dufek Massif are not included because of uncertainty about the need to apply a dip correction.

Measurements on middle Jurassic rocks from the Peninsula therefore pointed to the possibility of determining relative movement between this region and East Antarctica, at the same time helping to confirm or disprove the validity of the various Gondwana reconstructions that have been proposed.

The E coast appeared to be a suitable locality for study. There is evidence suggesting a westerly migration of igneous activity with time across the Peninsula (Saunders *et al.* 1982), hence reducing the likelihood of thermal resetting of the remanence of earlier

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TABLE 1: *Jurassic palaeomagnetic data for East Antarctica*

Location	*Source*	*Sites (samples)*	*Dec.*	*Inc.*	*^{a}95*	*k*	*Lat.* °S	*Long.* °E	*dp*	*dm*
Whichaway Nunatacks	Blundell 1966	7 (8)	64	68	14	21	54	224	12,	10
S Victoria Land	Bull *et al.* 1962	46 (83)	250	−68	3	63	45	220	4,	5
Victoria Land	Briden & Oliver 1963	9 (13)	244	−75	11	18	59	221	18,	20
Ferrar Glacier	Turnbull 1959	5 (57)	255	−76	3	714	58	218		5
Dronning Maud Land (Dykes)	Lovlie 1979	25 (109)	31	−59	4	47	54	208	4.8,	6.4

plutons in the E by later intrusions. Some of the older bodies in the eastern part of the Peninsula had already been dated as Jurassic (Rex 1976; Pankhurst 1982).

New palaeomagnetic data

Sampling for the palaeomagnetic studies was carried out during the austral summer of 1977–78 in two areas on the E coast of the Antarctic Peninsula (Fig. 1). The geology of these areas had been described by Stubbs (1968) and Marsh (1968), and a geochronological study has been made by Pankhurst (1982). On Cole Peninsula, the more southerly of the two areas, the granodiorite body of Mt. Hayes and the granite body of Mt. Denuce intrude older plutonic rocks. The

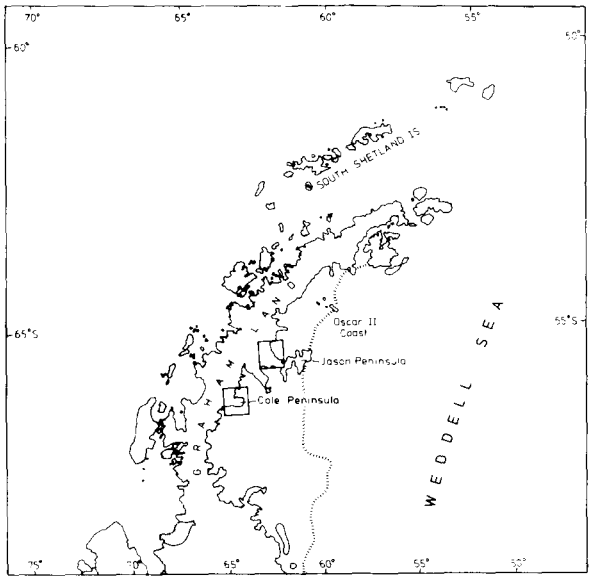


FIG. 1. The Antarctic Peninsula showing the palaeomagnetic sampling areas (boxed).

Rb-Sr isochron gave an age of 178 ± 2 Ma for the Mount Denuce granite, a comparable but less precise date being obtained for the Mt. Hayes body (R. J. Pankhurst, pers. comm.). A suite of acid to basic dykes intrudes the older plutonic rocks (dated at 209 ± 3 Ma) but is absent from the younger Mt. Denuce and Mt. Hayes bodies. An age of 174 ± 5 Ma for these dykes suggests they are contemporaneous with the younger plutons. Of the rock sampled from Cole Peninsula only the Mt. Hayes granodiorite and the dykes gave reliable palaeomagnetic results.

Rex (1976) reported dates from volcanic rocks from Jason Peninsula. Further ages of 157 ± 3 and 161 ± 3 (R. J. Pankhurst, pers. comm.) tend to confirm a middle Jurassic age. Two widely separated lava units provided consistent palaeomagnetic data.

In the northern area of Oscar II Coast samples were taken from the Laputa Nunataks granodiorite, the Mt. Fritsche granodiorite, the Bildad Peak tonalite, granodiorite and adamellite, and from the lava flows of Gulliver Nunatak. All failed to give consistent or reliable palaeomagnetic data.

Collection sites were well spaced with 4 or more samples taken at each site from plutonic bodies and 3 or more taken at a single site from each dyke.

Specimens from all samples were subjected to AF demagnetization and a selection of samples thermally demagnetized. AF demagnetization was normally carried out in steps of 100 gauss up to 500 gauss, though higher fields were occasionally used. Steps of 100°C were used for thermal demagnetization, normally up to a maximum of 400°C but occasionally to 600°C. The information obtained from these pilot samples was plotted on Zijderveld projections, examples of which are given in Fig. 2, enabling assessments to be made of the routine procedures to be adopted for the treatment of other samples. Fields used were those giving stable end points or minimum dispersion of remanence directions. When combining specimens to sample level, samples to site level, and in the case of plutonic bodies, sites to unit mean, the criterion adopted for acceptance was a circle of confidence of

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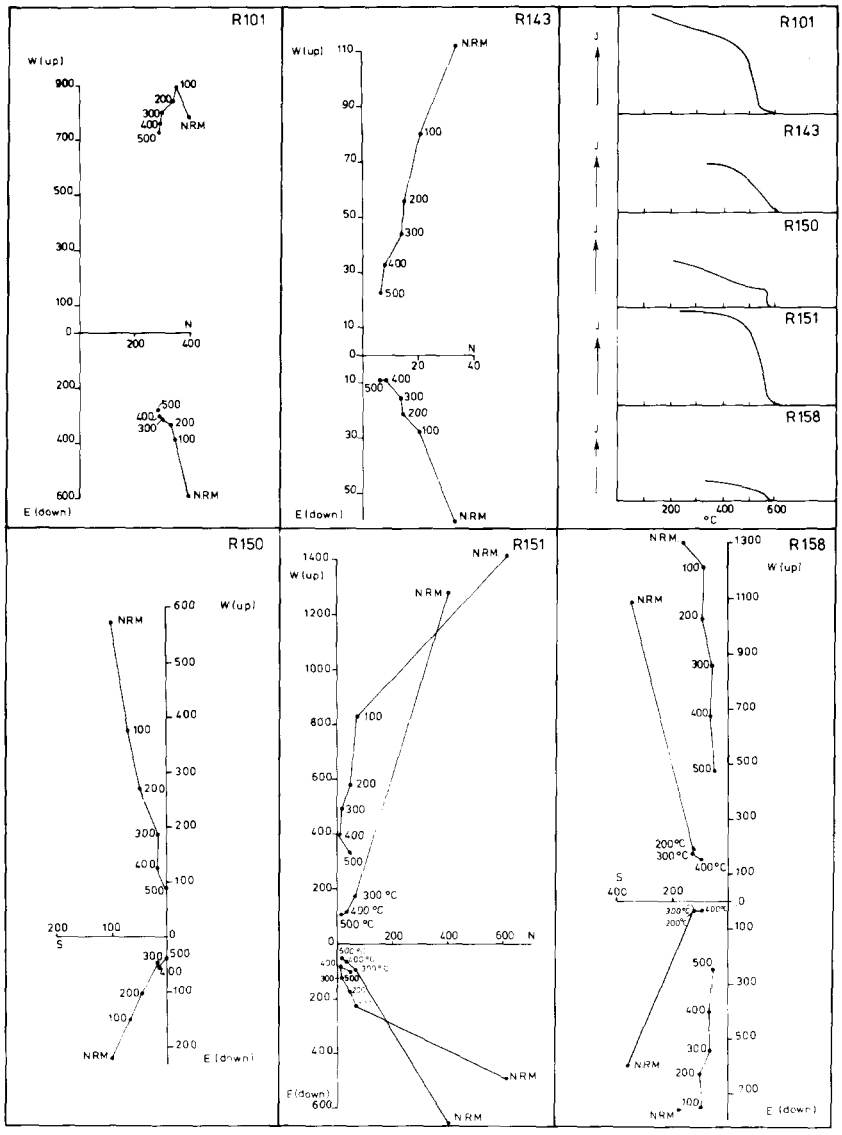


FIG. 2. Partial thermal and AF demagnetization experiments for specimens of dykes from the northern [R143] and the southern [R150, R151] Cole Peninsula, for the Mt. Hayes granodiorite [R158], and for a lava from Jason Peninsula [R101]. Curie balance curves for these specimens are also shown. All remanence directions are 'up' and E. Units are 10^{-6} emu cc^{-1}.

<25° at the 95% confidence level. In general a stable TRM component was isolated at fields of 200 to 300 gauss, whilst determination of bulk oxide composition indicated a single homogeneous magnetic component with a Curie temperature close to that of magnetite (Fig. 2).

A further statistical test was carried out on the dykes from the northern and southern areas of Cole Peninsula. The dykes dip at widely varying angles, averaging 74°. On the somewhat unlikely assumption that they were all intruded vertically and became magnetized before rotation the AF and thermally demagnetized dyke mean directions were corrected to the vertical, and sets of overall mean directions calculated. None of these showed any improvement in the dispersion parameter. No dip corrections were, therefore, applied to these or any of the other measurements.

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TABLE 2: *Palaeomagnetic and unit mean data from the E coast of the Antarctic Peninsula.* Data are for all statistically acceptable sites. AF demagnetized specimens only included

Locality	Site	Susceptibility ($\times 10^{-3}$ G/oe)	N.R.M. ($\times 10^{-6}$ e.m.u./cc) Average	N.R.M. Range	Median demag. field (gauss)	Samples N	Dec.	Inc.	α_{95}	k
Cole Peninsula Group 1	R138	1.41	143	24–316	≥600	4	117	−44	5.6	266
	R140 5–7	91.2	1274	950–1515	≤100	3	70	−70	16.0	60
	R142	6.46	232	105–415	200–400	4	76	−71	12.6	53
	R143 2–4	0.96	52.4	3–132	≈200	3	43	−68	11.3	120
	R143 6–8	79.4	1458	1131–1882	<100	3	43	−67	10.3	126
Cole Peninsula Group 2	R150 1–3	0.96	130	14–535	100–300	3	43	−75	8.4	212
	R150 4–6	6.6	497	103–891	200–300	3	107	−71	15.9	61
	R150 7–9	36.33	510	373–702	100–200	3	96	−72	8.8	197
	R150 10–12	28.3	121	16–402	≈200	3	86	−80	18.2	47
	R151 1–3	49.0	1939	1129–3397	≈200	3	87	−72	10.0	152
	R151 4–6	60.2	1296	896–1509	200–300	3	79	−78	12.6	96
	R152	47.9	836	540–1166	300–400	3	108	−66	11.6	114
	R153	55.0	934	443–1526	100–200	3	105	−69	3.6	1141
	R154 1–4	50.1	1439	1183–1758	200–300	4	118	−78	2.7	1082
	R154 5–7	30.2	192	127–227	≈200	3	102	−79	10.0	126
	R155	6.17	136	46–354	>300	3	127	−62	18.4	46
	R156 1–3	47.9	1731	22–3291	≈400	3	115	−72	7.2	267
	R156 4–6	2.75	194	22–624	100–200	3	102	−69	24.7	26
Mt. Hayes	R157	33.1	281	92–475	<200	4	66	−75	16.3	33
	R158	39.8	1013	367–1520	>400	4	90	−68	22.1	18
	R159	33.1	308	119–529	≤100	2	99	−67	—	—
	R160	26.3	335	130–716	≈100	4	104	−73	13.2	49
Jason Peninsula	R101	—	1377	273–2433	>500	4	42	−72	7.3	158
	R102	—	264	73–918	≥300	7	147	−61	8.3	53

TABLE 3: *Mean magnetization directions and palaeopoles for Jurassic rocks from the E coast of the Antarctic Peninsula*

	Site Lat °E	Site Long °S	Sites (Samples)	Dec.	Inc.	α_{95}	Palaeopole Lat °S	Palaeopole Long °E
1. Dykes of northern Cole Peninsula (Group 1)	296.0	66.8	5(17)	85	−66	22	−45	226
2. Dykes of southern Cole Peninsula (Group 2)	296.0	66.8	13(40)	91	−71	7	−49	236
3. Mt. Hayes Granodiorite	296.0	66.8	4(14)	91	−71	7.2	−49	236
4. Lavas of Jason Peninsula	298.5	66.3	2(11)	110	−75	—	−48	255
					Mean palaeopole:		−48	238 ± 9.5

Table 2 gives the relevant palaeomagnetic data for all statistically acceptable sites, magnetization directions being plotted on a stereographic projection in Fig. 3. Mean magnetization directions and palaeopoles are given in Table 3. The mean palaeopole for the Peninsula was obtained by averaging mean palaeopoles for the two groups of dykes, the Mt. Hayes granodiorite and the Jason Peninsula lavas, to give a position at 238°E 48°S.

Palaeomagnetic test of Gondwana reconstructions

Before attempting to use the new Jurassic palaeomagnetic data to locate the Antarctic Peninsula relative to the remainder of Gondwana it is necessary to decide on an initial shape for the supercontinent since this provides an additional spatial constraint. We have tested various reconstructions that have been

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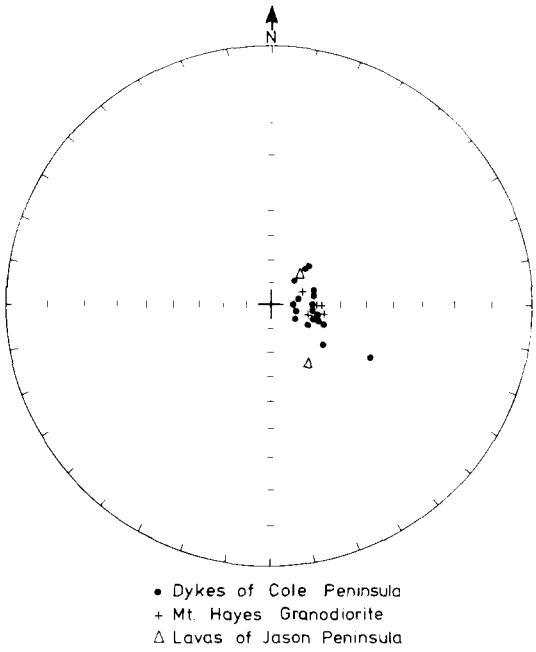


FIG. 3. Stereographic projection showing mean directions for the demagnetized samples of Table 2. North seeking poles plotted in upper hemisphere.

advanced against the very considerable body of Mesozoic palaeomagnetic data now available for Gondwana. Embleton *et al.* (1980) carried out a similar but less rigorous analysis for 5 age zones from the mid-Cambrian to the Jurassic, on the basis of which they accept a slightly modified Smith & Hallam (1970) reconstruction (Griffiths 1974). Table 4 is a compilation of Jurassic and Triassic palaeopole data for the Gondwana continents, this being divided into 4 age ranges. The palaeopole data for each of these continental units have been rotated accordingly for each of the 4 reconstructions given in Fig. 4, modified from Smith & Hallam (1970), Barron *et al.* (1978), Norton & Sclater (1979) and Powell *et al.* (1980). Common to all models is the fit of Africa and South America, for which the geometric and sea-floor spreading evidence is very good. It does, however, result in poor agreement of the palaeopole data. As shown in Table 4 comparison of the palaeopole data for each of the 4 age ranges with and without the South American data indicate an obvious deterioration of the precision parameter k when the South American data are included. [Possible reasons for this have been suggested by Palmer *et al.* (1980).] The comparisons have therefore been carried out excluding the South American data. For the models of Norton & Sclater and modified Smith & Hallam, mean palaeopoles for Gondwana have a tighter grouping. This is indicated by smaller circles of confidence and higher precision parameters for the Upper and Middle Jurassic, the Lower Jurassic and the Upper Triassic but not for the Middle and Lower Triassic, for which times the data are much more sparse. There is little difference in the statistics for the two models and, of the two, we have adopted that of Norton & Sclater. Combining all Jurassic data for Gondwana with the exclusion of South America and carrying out the same operation for the Triassic, two mean palaeopoles are obtained (Table 5).

A value of 3.31 for the 'comparison of mean directions' statistic (Watson 1956) suggests that the two palaeopoles are significantly different at the 95% level, though the variance ratio is 2.6, when preferably it should be unity, which would signify similar precision limits for the two populations.

Comparing the Jurassic mean palaeopoles for Australian and East Antarctic data strongly suggests that the magnetic data here are not inconsistent with the reconstruction, despite the fact that the comparison is made using some possibly late Jurassic data, i.e. post Gondwana separation. The data are, however, weighted heavily towards the early Jurassic. If the mean African Jurassic palaeopole is now compared with the combined East Antarctica/Australian mean pole for the various models, a mean direction statistic of 1.66 is obtained for the Norton & Sclater model, one of 23.9 for that of Barron *et al.* and 9.3 for Powell *et al.*, endorsing the rejection of the two latter models at the 99% probability level.

Jurassic position of the Antarctic Peninsula

Adoption of the Norton & Sclater reconstruction of Gondwana implies that the Peninsula had a different position relative to East Antarctica in the Jurassic. As yet no firm evidence has been published to indicate the precise relationships of the two areas but geometrical and palaeomagnetic constraints can be used to limit the possibilities, as we demonstrate below.

Little is yet known about the structure of the area between the Weddell Sea and the Ross Sea but the suggestion has been made that it consists of a number of microplates. Geological and geophysical evidence certainly indicate the likelihood of there being an Ellsworth microplate (Schopf 1969; Watts & Bramall 1981; Jankowsky & Drewry 1981) and the continental area of the head of the Weddell Sea may be another, the Filchner microplate (F in Fig. 6). Tentatively, therefore, the Peninsula is also taken to be a separate entity with a boundary defined by the present 2000 m contour for the E and a conjectural boundary to the S based on sub-ice topography. In the middle Jurassic the Peninsula was no doubt narrower than now and the configuration of the continental areas at its

TABLE 4: *Mean palaeopoles for Gondwana for 4 time intervals in the Mesozoic calculated for 4 different reconstructions.*

	Upper & Middle Jurassic				*Lower Jurassic*				*Upper Triassic*				*Middle & Lower Triassic*			
Numbers of palaeopoles per continent																
Africa		—				10				3				6		
South America		4				2				3				—		
East Antarctica		4				1				1				—		
India		1				1				3				—		
Madagascar		1				1				1				—		
Australia		1				4				—				2		
	Lat. °S	*Long. °E*	A_{95}	*k*	*Lat. °S*	*Long. °E*	A_{95}	*k*	*Lat. °S*	*Long. °E*	A_{95}	*k*	*Lat. °S*	*Long. °E*	A_{95}	*k*
Norton & Sclater	−59	219	7	70	−57	219	4	88	−55	229	7.5	55	−54	230	12	21
(1979)	*−61*	*214*	*6*	*48*	*−58*	*219*	*4*	*79*	*−60*	*230*	*9*	*28*				
Smith & Hallam (1970) modified after Griffiths	−59	219	7	69	−55	216	4	91	−53	227	7	60	−52	226	13	19
(1974)	*−60*	*212*	*7*	*44*	*−56*	*216*	*4*	*83*	*−58*	*227*	*8*	*30*				
Barron *et al.*	−60	220	7	66	−66	205	5	44	−62	229	11	27	−67	214	8	45
(1978)	*−67*	*205*	*10*	*20*	*−68*	*203*	*5*	*400*	*−70*	*226*	*12*	*15*				
Powell *et al.*	−61	218	8	58	−64	215	4	66	−60	228	9	38	−63	230	10	31
(1980)	*−66*	*208*	*8*	*31*	*−65*	*215*	*4*	*57*	*−67*	*227*	*10*	*200*				

Figures in italics are calculated values which include South American data. The mean palaeopoles in the table are calculated from palaeopoles in the following publications: Irving & Stott (1963); McElhinny (1968–1972); McElhinny & Cowley (1977); Daly & Pozzi (1976); Palmer *et al.* (1980); Robertson & Hastie (1962); Schmidt (1976); Embleton & McElhinny (1975); Klootwijk & Bingham (1980).

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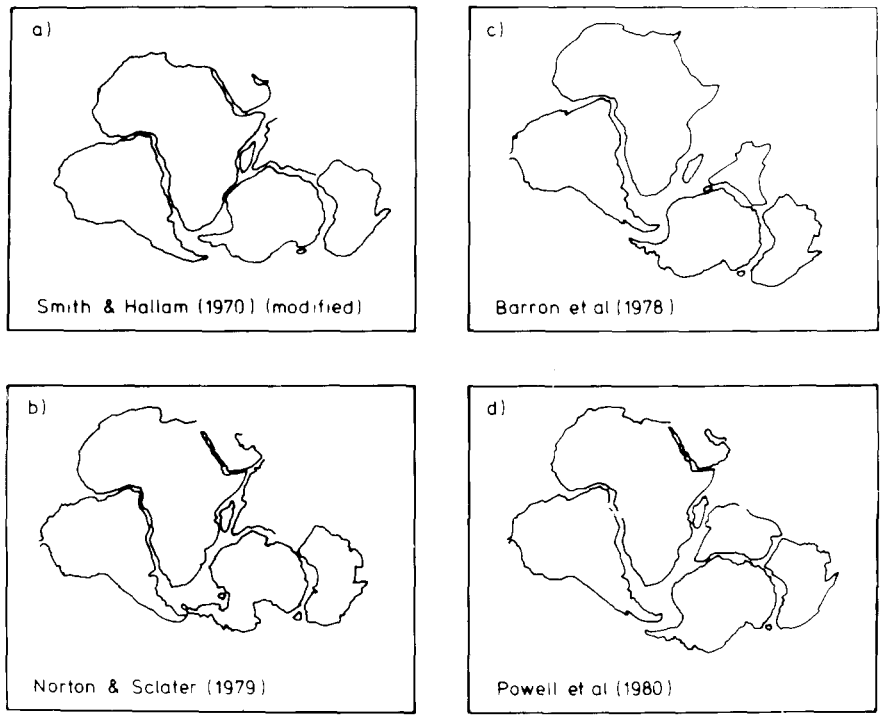


FIG. 4. Four reconstructions proposed for Gondwana.

northern end was undoubtedly different. Bransfield Strait had not opened and the position of the South Orkney Is block relative to the Peninsula was probably different. However, Jurassic ages from the northern Antarctic Peninsula (Rex 1976) indicate that its longitudinal extension was not substantially less at that time.

If the Jurassic palaeomagnetic poles for East Antarctica and the Peninsula are made to coincide, then the latter is constrained to lie on a circle at a fixed distance from that pole and have a fixed orientation relative to a radius from the pole to the plate. Its position on the circumference has to be otherwise determined. In Fig. 5 the two palaeopoles have been made coincident and the Peninsula (present continental outlines but without the South Shetland Is and the South Orkney Is) has been rotated to fit between the Falkland Plateau and East Antarctica. The fit is a tight one but, as pointed out earlier, the size and shape of the Peninsula as drawn was no doubt somewhat, though probably not greatly, different in the Jurassic.

The continental margin of South America and the Falkland Plateau may have also extended since the Jurassic, a consequence of active margin processes, including the formation and subsequent closing of the Sarmiento basin (Dalziel *et al.* 1974). The northern tip

TABLE 5: *Mean palaeopoles calculated for a Gondwana reconstruction according to Norton & Sclater (1979).* Palaeopole data used is that listed in Table 4

Palaeopole data	*Age*	*Number of palaeopoles*	*Long. °E*	*Lat. °S*	A_{95}	*k*
Gondwana	(J)	24	219	−57	3.2	85
Gondwana	(Tr)	16	230	−55	6.5	33
Gondwana	(J/Tr)	40	223	−56	3.2	49
(Gondwana incl. South America		49	222	−59	3.3	38)
East Antarctica	(J)	5	217	−54	6.0	159
Australia	(J)	5	210	−57	9.2	69
East Antarctica/Australia	(J)	10	214	−55	4.9	96
Africa	(J)	10	222	−56	4.9	97
Africa/(East Antarctica/Australia)	(J)	20	218	−56	3.3	94

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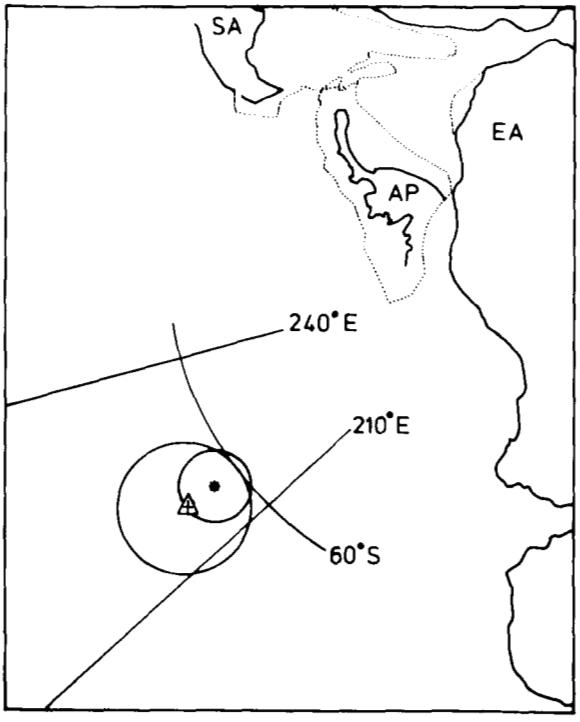


FIG. 5. A Jurassic position for the Antarctic Peninsula consistent with the palaeomagnetic data. The palaeopole for the Peninsula has been made coincident with that for East Antarctica. The reconstruction for the other continents is that of Norton & Sclater (1979). Coordinates used are relative to the present position of Antarctica.

Position of Jurassic palaeopoles:

East Antarctica	Δ
Antarctic Peninsula	+
Gondwana (excluding South America)	*

of the Peninsula lies close to the position it had at the time of opening of Drake Passage (Barker & Burrell 1977) and according to Norton & Sclater (1979) the relative positions of the Peninsula and southern South America need have changed little as far back as 115 Ma. The reconstruction leaves space for South Georgia close to South America and at the Pacific margin (Simpson & Griffiths 1982; Dalziel *et al.* 1974).

Permitting the Peninsula palaeopole to lie anywhere within the circle of confidence of the East Antarctic palaeopole allows some latitude in the positioning of the Peninsula but reconstructions such as those of De Wit (1977) in which the present Weddell Sea margin of the Peninsula is made to lie along the southern edge of the Falkland Plateau are unacceptable. A much more easterly position for the northern end, as preferred by Dalziel (1982), is also inconsistent with the palaeomagnetic data.

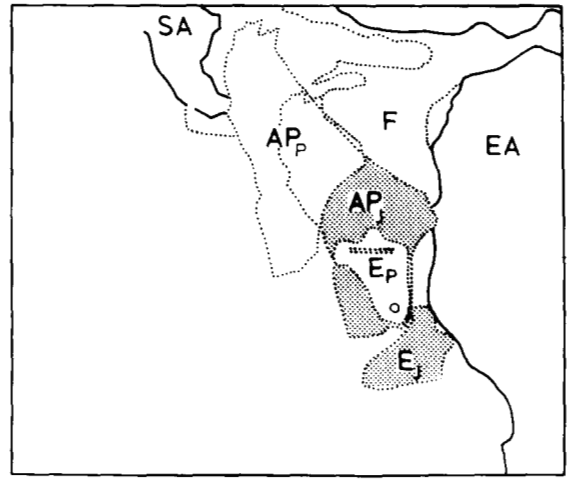


FIG. 6. Jurassic (AP_J) and present (AP_P) position of the Antarctic Peninsula and Ellsworth Mts. [E_J, E_P] microplate. Filchner microplate shown as F. Ellsworth Mts., ::::; Whitmore Mts., ○.

Movement of the Peninsula relative to East Antarctica from the position shown in Fig. 5 to its present position is equivalent to a clockwise rotation of 15° about a pole located at 65°S 0°E. If the Ellsworth and Whitmore Mts. are part of one microplate (Jankowski & Drewry 1981) the reconstruction favours an original position for this plate below the base of the Antarctic Peninsula, as shown in Fig. 6, rather than one close to the Pensacola Mts. as suggested by Schopf (1969) and Watts & Bramall (1981). Without relaxing the palaeomagnetic constraint unduly or accepting a much more complex movement for the Antarctic Peninsula than is proposed here, considerations of space make it impossible to rotate the Ellsworth block from the vicinity of the Pensacola Mts. to its present day position. If it originally lay close to the southern edge of the Peninsula as shown in Fig. 6 then only minimal movement, involving an anticlockwise rotation of about 90° about a near pole consistent with the palaeomagnetic data of Watts & Bramall, is needed to bring the block into its present position relative to the Peninsula and East Antarctica. This proposed Jurassic position for the Ellsworth block places the Whitmore Mts. (Webers *et al.* 1982) close to what must have been the Pacific margin, which is consistent with the fact that they contain calc-alkaline plutons dated as Middle Jurassic. It is interesting to note that the movement of the Ellsworth block relative to the Antarctic Peninsula could have created the Byrd Basin (Fig. 7). The 90° anticlockwise rotation is in fact greater than that required to produce the basin. It is necessary therefore to presume that additional adjustment to other small continental blocks has occurred or that the rotation of the Ellsworth plate was less than the palaeomagnetic data appear to indicate.

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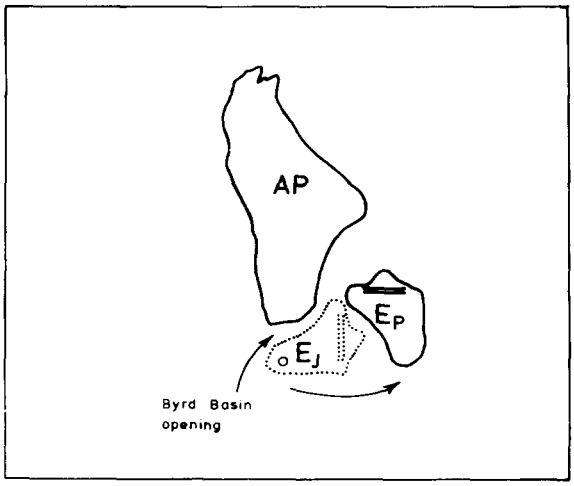


FIG. 7. Jurassic (E_J) and present (E_P) positions of the Ellsworth microplate relative to the Antarctic Peninsula (AP), illustrating the formation of the Byrd Basin.

Palaeomagnetic data from the Peninsula for the Cretaceous are capable of more than one interpretation and it is not therefore possible to date the movement of the Peninsula relative to East Antarctica, nor to determine its path. If the proposed initial position is correct (Fig. 7), a quite simple movement is suggested and it is possible that initially the Peninsula moved as part of the South American plate, at some stage decoupled from East Antarctica along a transform fault parallel or close to the present Weddell Sea margin. These ideas have something in common with models suggested by Barker & Griffiths (1977) but not enough is yet known to enable a full and consistent account to be given of the evolution of the area.

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